

scRAW

Base commune :

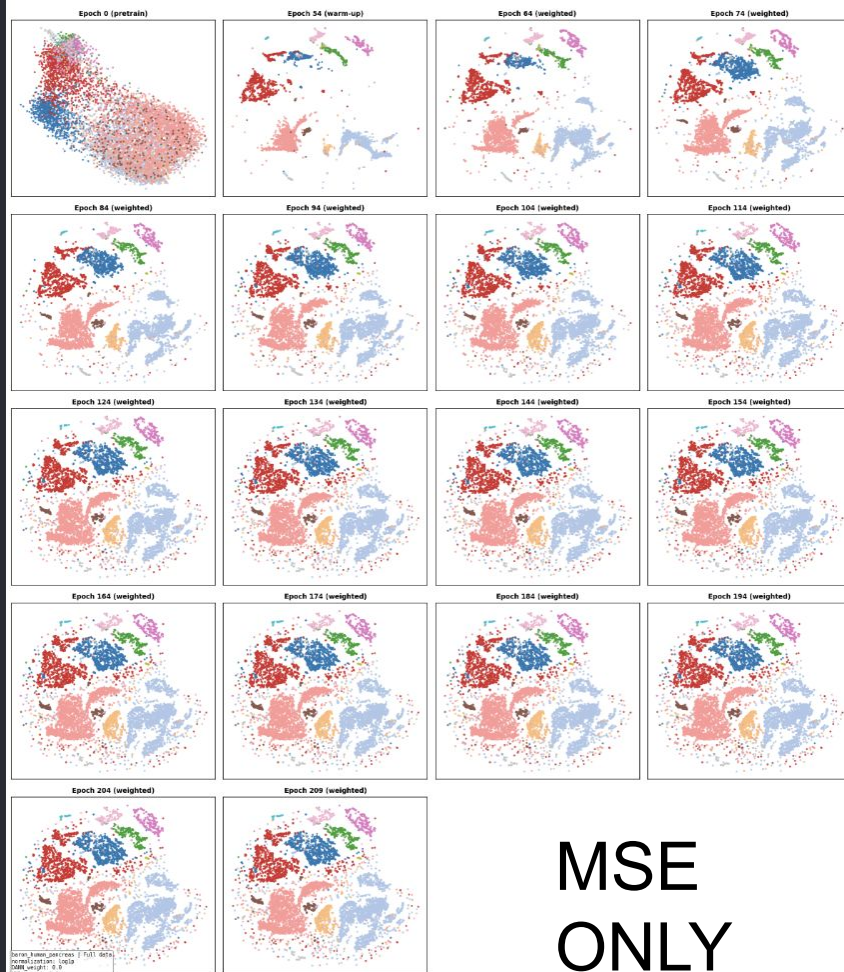
hidden_layers=512,256,128,
z_dim=192,
dropout=0.3,
batch_size=192,
lr=0.002,
reconstruction mse,
masking_rate=0.15,
pseudo-labels leiden,
clustering final hdbscan,
pondération additive cluster+density,
dynamic_weight_momentum=0.85,
dynamic_weight_update_interval=10

Phase commune:

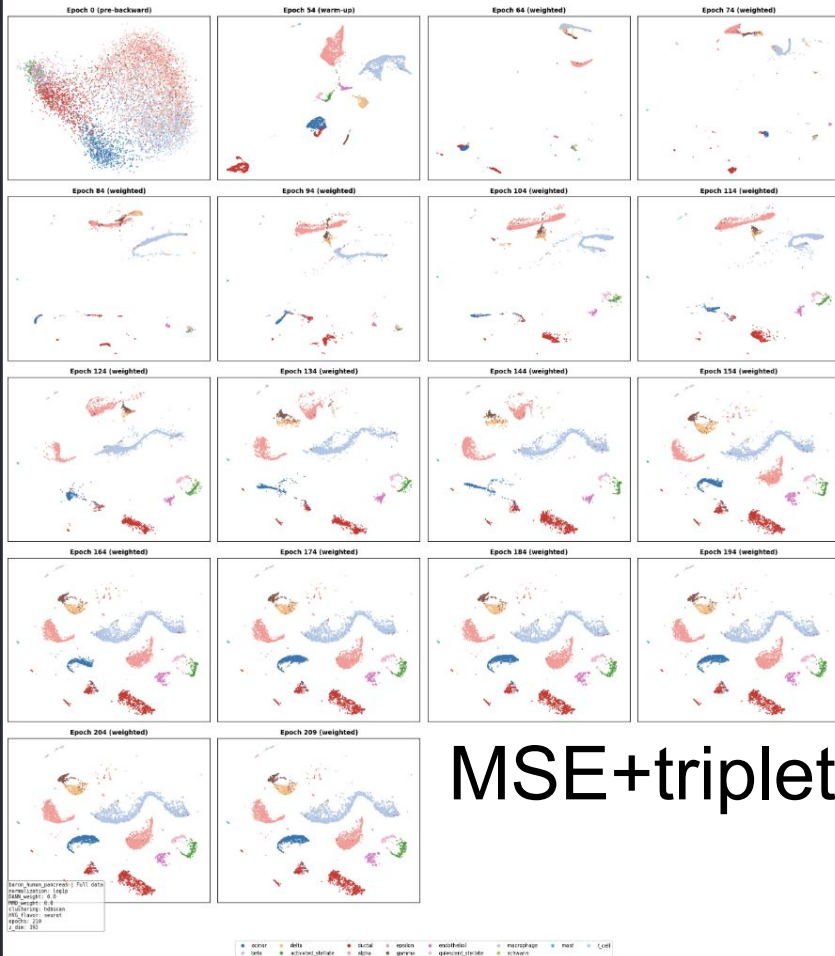
entraînement jusqu'à l'époque 54,
avec rare_triplet_weight=0 et
adversarial_batch_weight=0, puis
sauvegarde d'un checkpoint partagé

Reprise des ablations à partir de l'
époque 55:
total 210 époques

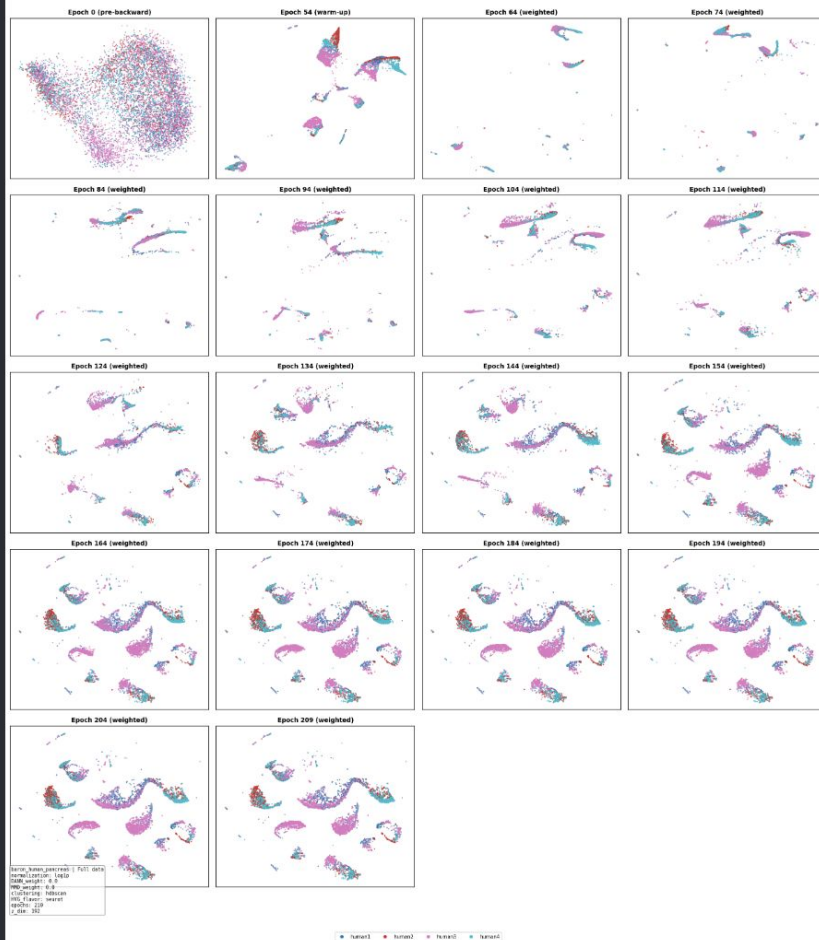
A	B	C	D	E	F	G	H	I	J
rank_by_ari	run_name	seed	ARI	ACC	BalancedACC	F1_Macro	RareACC	NMI	Silhouette
1	complet_seed_42	42	0.9364	0.9459	0.8943	0.7976	0.9658	0.9065	0.7253
2	run_ablation_dann_mse_triplet	42	0.9323	0.9315	0.6884	0.6548	0.7939	0.8999	0.7289
3	run_ablation_no_momentum	42	0.9279	0.9299	0.7562	0.6975	0.8004	0.8964	0.6762
4	run_ablation_no_cluster_weight	42	0.9256	0.9297	0.7544	0.6969	0.8004	0.8949	0.7563
5	seed_seed_139	139	0.9124	0.9406	0.8930	0.7875	0.9621	0.8905	0.5953
6	run_ablation_single_update	42	0.8814	0.9030	0.7461	0.6863	0.7902	0.8636	0.6086
7	complet_seed_236	236	0.8718	0.9041	0.8231	0.7550	0.8189	0.8612	0.7526
8	run_ablation_triplet_mse	42	0.7943	0.7797	0.5849	0.5599	0.5656	0.8394	0.6955
9	complet_ablation_momentum0_minimal	42	0.7791	0.8553	0.7575	0.7054	0.7348	0.8298	0.6794
10	run_ablation_mse_only	42	0.4786	0.5864	0.6948	0.3852	0.6922	0.6830	0.1493
11	run_ablation_dann_mse	42	0.2308	0.4312	0.3650	0.3660	0.4381	0.4577	0.1253

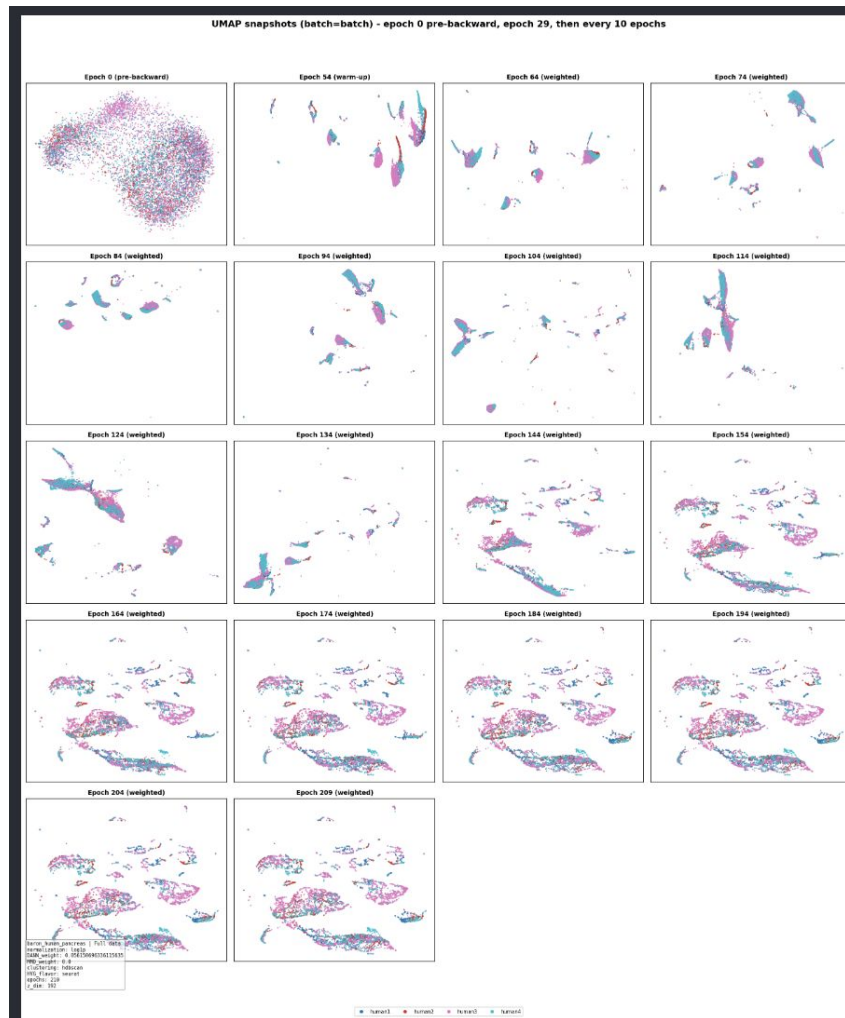
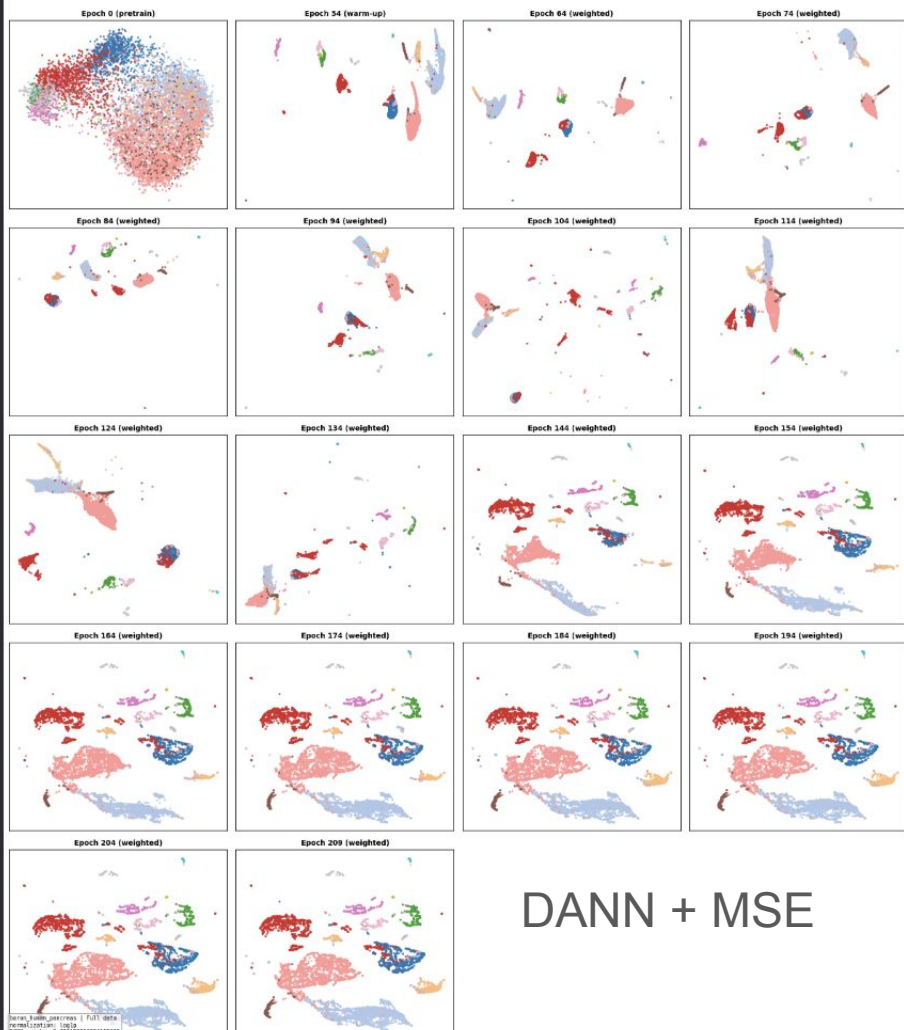


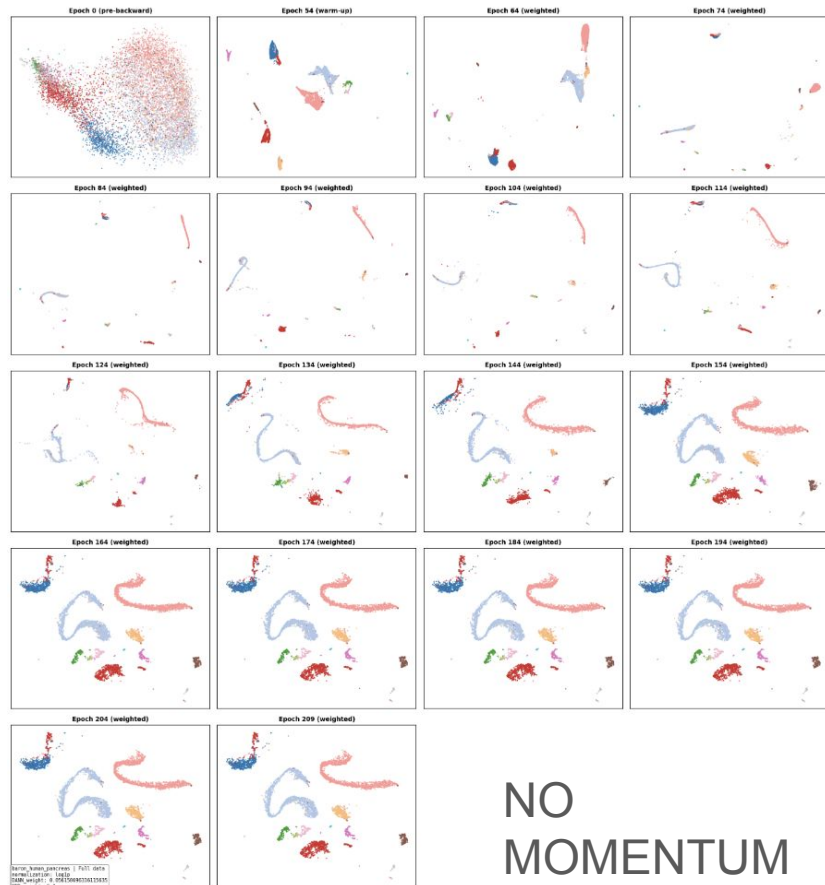
UMAP snapshots (labels) - epoch 0 pre-backward, epoch 29, then every 10 epochs



UMAP snapshots (batch=batch) - epoch 0 pre-backward, epoch 29, then every 10 epochs



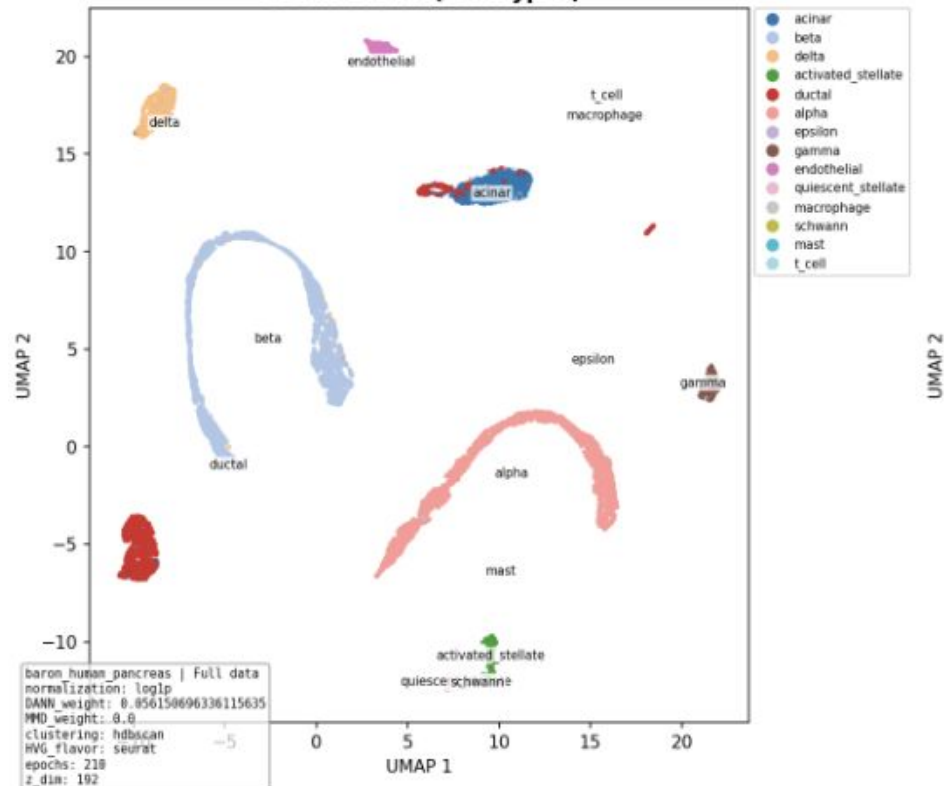




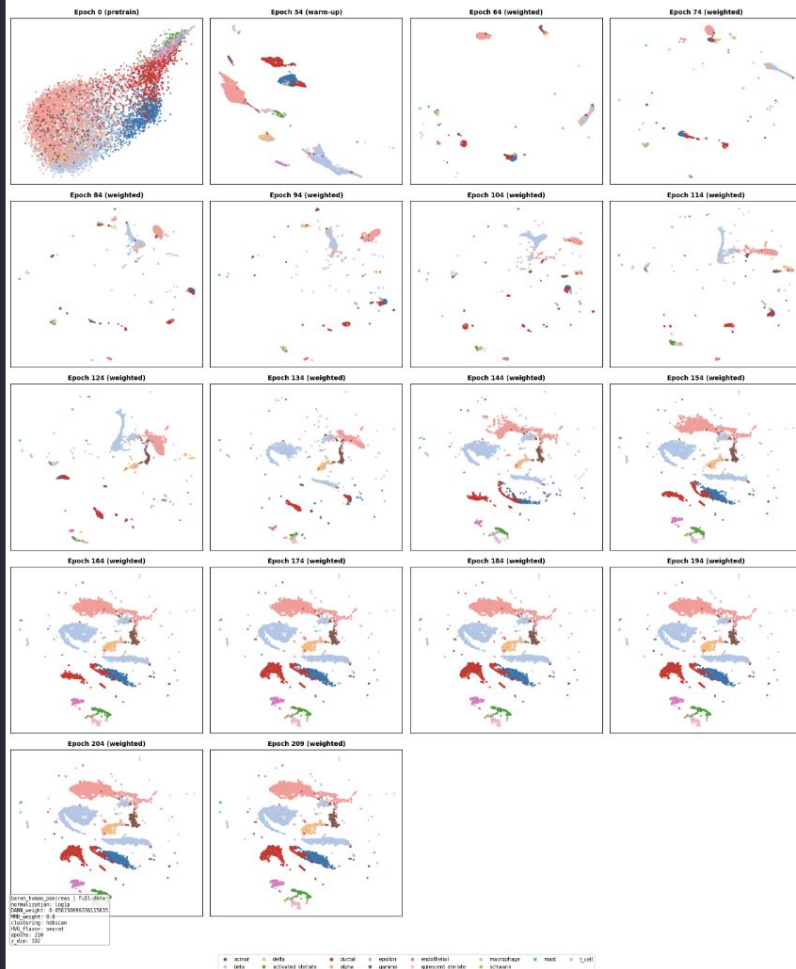
NO
MOMENTUM

acinar beta delta ductal epsilon gamma quiescent stellate macrophage mast t_cell
delta activated_stellate alpha schwann

Ground Truth (Cell Types)



UMAP Evolution - scraw (Ground Truth | shared-UMAP (precomputed) | 18/18 snapshots)



Ground Truth (Cell Types)

